



Inspiring Excellence

Assignment-03

Combinational Circuits

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Course: CSE260

Section: 17B

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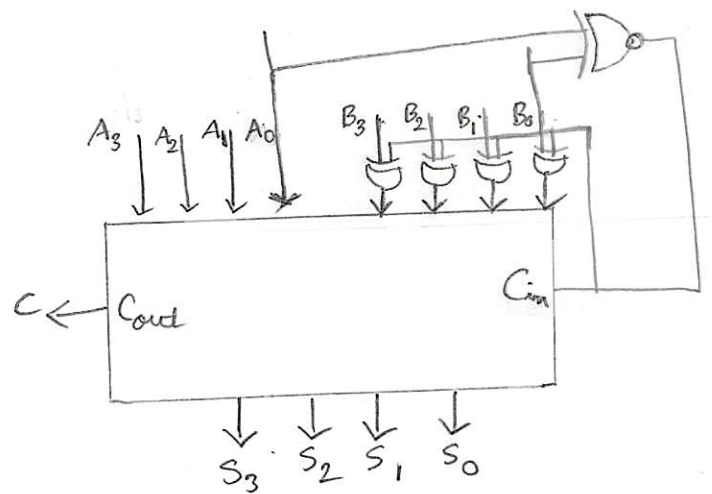
Q.1

	B_3	B_2	B_1	B_0
	A_3	A_2	A_1	A_0
0 →	0	0	0	0
1 →	0	0	0	1
2 →	0	0	1	0
3 →	0	0	1	1
4 →	0	1	0	0
5 →	0	1	0	1
6 →	0	1	1	0
7 →	0	1	1	1
8 →	1	0	0	0
9 →	1	0	0	1
10 →	1	0	1	0
11 →	1	0	1	1
12 →	1	1	0	0
13 →	1	1	0	1
14 →	1	1	1	0
15 →	1	1	1	1

Even when $LSB = 0$

Odd when $LSB = 1$

A_0	B_0	Add = 0 Subtract = 1
0	0	1
0	1	0
1	0	0
1	1	1



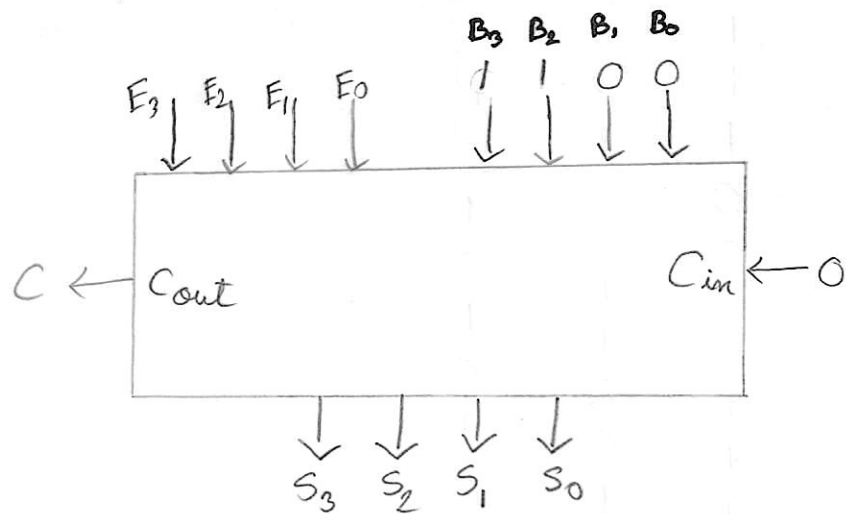
Q.2

Excess-4 : $E_3 E_2 E_1 E_0$

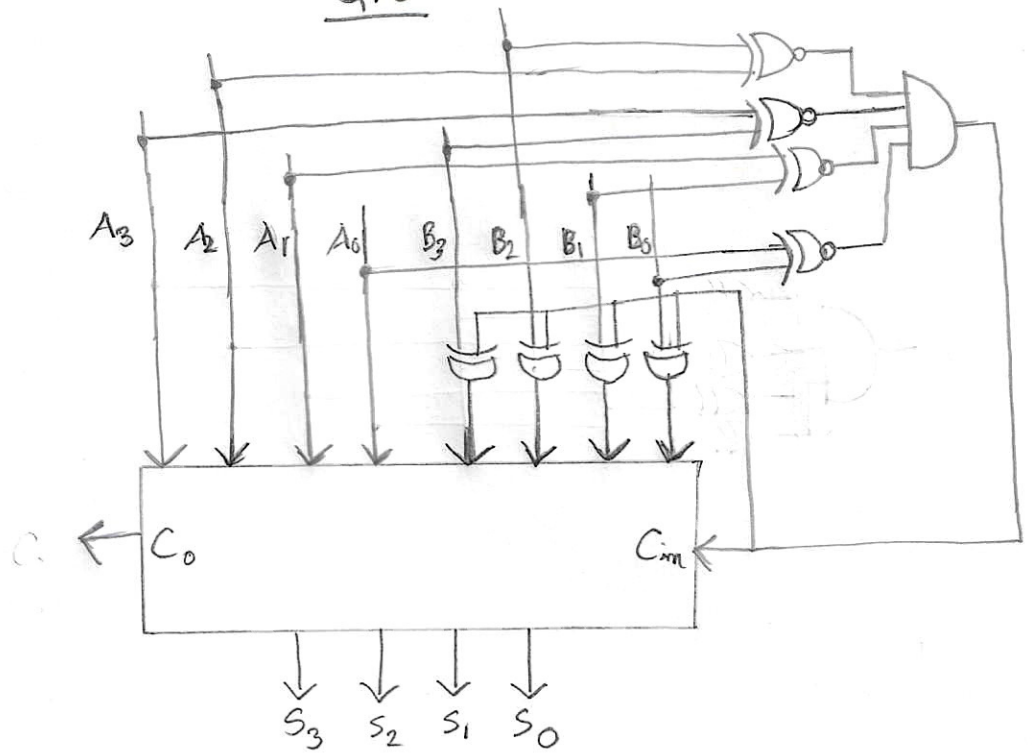
$4 \rightarrow (0100)_2$

1's complement : $\begin{array}{r} 1011 \\ +1 \end{array}$

2's complement : $\begin{array}{r} 1100 \\ B_3 B_2 B_1 B_0 \end{array}$



Q.3



ASSIGNMENT 3: [5+5+5]/3

Q1. Design a circuit diagram for the following system that takes two 4-bit binary numbers [A and B] as inputs and outputs in the following fashion:

- If both inputs are odd or both inputs are even, the output should be $A-B$
- If one input is odd and the other input is even, the output should be $A+B$

Q2. With 4-bit Parallel Adder, design a converter where if you enter an Excess-4 number, it will give you back the BCD of it. For Example: if you enter 0101, you will get output as 0001.

Q3. Design a circuit diagram for the following system that takes two 4-bit binary numbers [A and B] as inputs and outputs in the following fashion:

- If both inputs are same, the output should be $A-B$
- Otherwise, the output should be $A+B$

Deadline : Aug 14, 2026 at 11 : 59 PM

In case of late Submissions, 10% marks will be deducted for every 12 hours of delay.

Submission Link: <https://forms.gle/WctaMdNG5Usoark4A>

$$\begin{array}{cccc} 1 & 0 & 1 & 0 \\ A_3 & A_2 & A_1 & A_0 \\ 1 & 0 & 1 & 0 \\ B_3 & B_2 & B_1 & B_0 \end{array}$$

0-0-0-0